

International Institute of Information Technology, Bangalore
CSE 102P Data Structures and Algorithms Lab- Sec A
Test1: 4 February 2026.

You are given three integers: n , m , and k . Define a function $F(i)$ as follows:

- For $i = 0, 1, 2, \dots, k$: $F(i) = i$
- For $i > k$: $F(i) = (F(i-1) + F(i-k)) \bmod m$

Given n , m , and k , compute $F(n)$.

Input Format

A single line containing three space-separated integers: n m k

Output Format

A single integer: $F(n) \bmod m$

Examples

Input	Output
5 1000 2	8
10 7 3	6
0 100 5	0
3 100 5	3
6 100 3	9

Explanation of Examples

Example 1: $n = 5$, $m = 1000$, $k = 2$

$$F(0) = 0$$

$$F(1) = 1$$

$$F(2) = 2$$

$$F(3) = F(2) + F(1) = 2 + 1 = 3$$

$$F(4) = F(3) + F(2) = 3 + 2 = 5$$

$$F(5) = F(4) + F(3) = 5 + 3 = 8$$

Thus, $F(5) = 8$.

Example 2: $n = 10$, $m = 7$, $k = 3$

$$F(0) = 0, F(1) = 1, F(2) = 2, F(3) = 3$$

$$F(4) = F(3) + F(1) = 3 + 1 = 4 \pmod{7}$$

$$F(5) = F(4) + F(2) = 4 + 2 = 6 \pmod{7}$$

$$F(6) = F(5) + F(3) = 6 + 3 = 9 \equiv 2 \pmod{7}$$

$$F(7) = F(6) + F(4) = 2 + 4 = 6 \pmod{7}$$

$$F(8) = F(7) + F(5) = 6 + 6 = 12 \equiv 5 \pmod{7}$$

$$F(9) = F(8) + F(6) = 5 + 2 = 7 \equiv 0 \pmod{7}$$

$$F(10) = F(9) + F(7) = 0 + 6 = 6 \pmod{7}$$

Thus, $F(10) = 6$.

Example 3: $n = 0$, $m = 100$, $k = 5$

Since $n = 0 \leq k$, $F(0) = 0$.

Example 4: $n = 3$, $m = 100$, $k = 5$

Since $n = 3 \leq k$, $F(3) = 3$.

Example 5: $n = 6$, $m = 100$, $k = 3$

$$F(0) = 0, F(1) = 1, F(2) = 2, F(3) = 3$$

$$F(4) = F(3) + F(1) = 3 + 1 = 4$$

$$F(5) = F(4) + F(2) = 4 + 2 = 6$$

$$F(6) = F(5) + F(3) = 6 + 3 = 9$$

Thus, $F(6) = 9$.

Constraints

- $0 \leq n \leq 10^{18}$
- $1 < m \leq 10^5$
- $0 < k < 100$